EDC · PAPER

EDC practice test: marking guide

Answers and worked solutions · Day 08 · Sat 31 Oct · pack EPT

Question-by-question marking guide

Question 1: multiple choice [5 marks]

Paper wording. Circle the correct answer in the following multiple-choice questions (I to V).

I. When a PN junction is formed, why does a depletion region develop near the junction?
a) Electrons and holes recombine near the junction, leaving behind ionized dopants
b) The semiconductor becomes electrically insulating everywhere
c) All free electrons move from the N-region into the P-region permanently
d) The applied voltage removes all charge carriers from the semiconductor

II. The small-signal or dynamic resistance of a diode is related to:
a) The slope of the I–V characteristic at the operating point
b) The total length of the diode leads
c) The reverse breakdown voltage only
d) The DC supply resistance only

III. The main advantage of full-wave rectification over half-wave rectification is that full-wave rectification:
a) Uses only the negative half of the input waveform
b) Uses both half-cycles of the input waveform
c) Completely eliminates ripple without a capacitor
d) Requires no semiconductor devices

IV. Why is the base region of a BJT made relatively thin?
a) To allow most injected carriers to cross the base and reach the collector
b) To prevent carriers from reaching the collector
c) To increase the base resistance as much as possible
d) To eliminate the depletion region at the collector-base junction

V. Why are DC and AC analyses of a transistor amplifier often performed separately?
a) A transistor can operate with either DC or AC but never both
b) DC analysis establishes the operating point, while AC analysis examines small variations around that point
c) AC analysis determines the transistor’s doping concentration
d) DC analysis is only needed when the transistor is damaged

Part	Answer	One-line reason
I	**(a)**	Carriers diffuse across and recombine near the junction, leaving immobile ionised donors (+, n side) and acceptors (−, p side). That bound charge is the depletion region.
II	**(a)**	$r_d = \Delta V_D / \Delta I_D = nV_T / I_D$, the inverse slope of the i–v curve at the Q-point.
III	**(b)**	It uses both half-cycles. That gives a higher DC output and twice the ripple frequency. It does *not* remove ripple without a capacitor, so (c) is wrong.
IV	**(a)**	Injected carriers cross the thin base before they recombine, so most reach the collector. That is why $\alpha \to 1$ and β is large.
V	**(b)**	DC analysis sets the Q-point. AC analysis linearises the device around that point (small-signal model).

This agrees with the Test 1 Reference.

Mark breakdown: 1 mark per correct letter. There are no part marks.

Most likely ways to lose marks:

- Changing (I) to (c) because “electrons move to the p side” sounds right. Diffusion stops when the field balances it, and it is not “all” electrons, “permanently”.
- Picking (c) in III because full-wave ripple is *smaller*. Smaller is not the same as eliminated.

Question 2: exponential model [5 marks]

Paper wording. Consider a silicon diode with its exponential model and $n = 1.5$. Find the change in the voltage across it when the diode operates at 27°C and the current changes from 0.1 mA to 10 mA.

Step 1: thermal voltage. $T = 273 + 27 = 300$ K.

$$V_T = \frac{kT}{q} = \frac{1.38 \times 10^{-23} \times 300}{1.6 \times 10^{-19}} = 25.875 \text{ mV}$$

Step 2: ratio form (removes the unknown I_S). For forward bias, $I_D \approx I_S e^{V_D/nV_T}$, so

$$\frac{I_2}{I_1} = e^{(V_2 - V_1)/nV_T} \Rightarrow \Delta V_D = nV_T \ln \frac{I_2}{I_1}$$

Step 3: substitute. $I_2/I_1 = 10/0.1 = 100$, and $\ln 100 = 4.605$.

$$\Delta V_D = 1.5 \times 25.875 \text{ mV} \times 4.605 = \mathbf{0.179 \text{ V } (179 \text{ mV})}$$

Sanity check: one decade costs $2.303\, nV_T = 89.4$ mV, so two decades cost 179 mV.

Python: 0.17874 V using T = 300 K. Using T = 300.15 K gives 0.17883 V, which is the same to 3 s.f. Using the rough $V_T = 25$ mV gives 0.173 V. That is acceptable only if you say you rounded V_T, and it may cost the "correct value" mark. This agrees with the Test 1 Reference (0.179 V).

Mark breakdown (5):

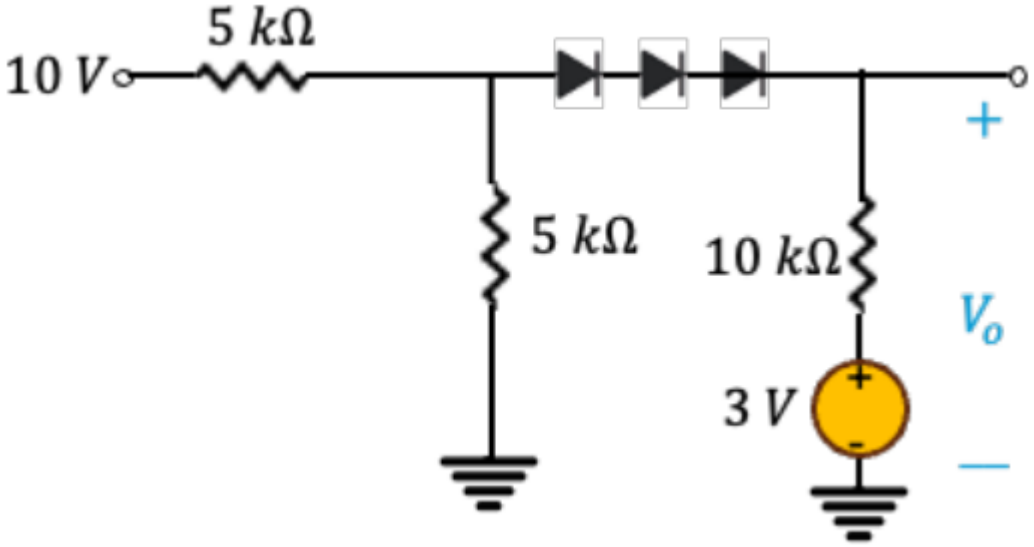
- 1: V_T computed at 300 K (25.9 mV), using the constants from the formula sheet.
- 2: the ratio derivation, showing that I_S cancels and arriving at $\Delta V = nV_T \ln(I_2/I_1)$.
- 1: correct substitution (n = 1.5, ratio 100).
- 1: final answer ≈ 0.179 V with units.

Most likely ways to lose marks:

- Dropping n (writing $V_T \ln 100 = 119$ mV), or using $\log_{10}$ instead of $\ln$ (which gives 77.6 mV).
- Using T = 27 instead of 300 K, or not showing the I_S cancellation, which costs the derivation marks.

Question 3: three series diodes, third PWL model [8 marks]

Paper wording. Consider the circuit below, which consists of the series combination of three identical diodes. Using the third piecewise linear model for the diodes with $V_D = 0.6$ V and $r_D = 0.20$ kΩ, answer the following questions.



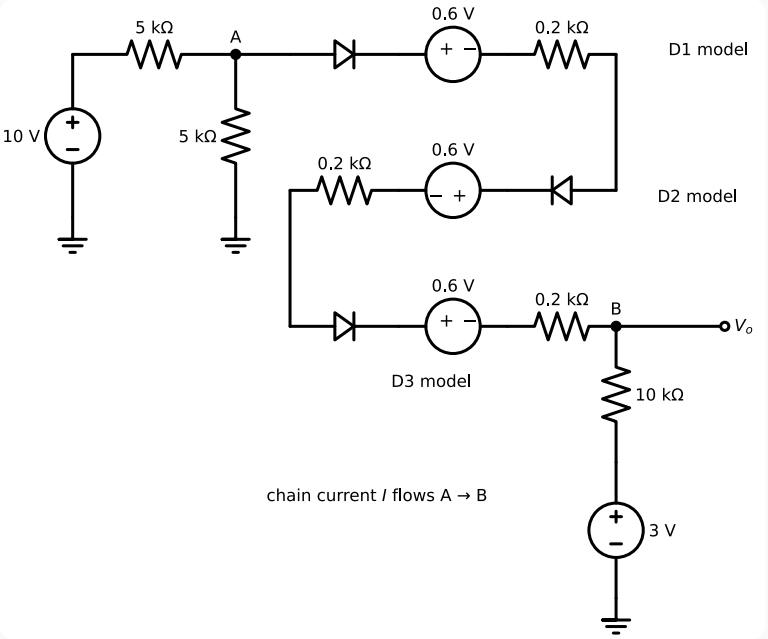
Q3 figure (from the practice-test paper)

Assume that all diodes are **conducting**.
(a) Redraw the entire circuit using the third piecewise linear equivalent circuit model of diodes. You do not need to simplify the circuit. [2 marks]
(b) Verify whether all three diodes are **conducting** under the given circuit conditions. [4 marks]
(c) Find the output voltage, V_o, as shown on the circuit. [2 marks]

READING THE FIGURE: 10 V source, then 5 kΩ to node A. Node A has 5 kΩ to ground. From A, three diodes in series (anodes towards A) run to node B = V_o. From B, 10 kΩ goes to the + terminal of a 3 V source whose − terminal is grounded.

(a) Redraw [2]. Replace *each* diode with an ideal diode in series with a 0.6 V source (+ towards A) and a 0.2 kΩ resistor. Draw all three separately, because the question says not to simplify:

10 V — 5 kΩ — **A** — [ideal D1, 0.6 V, 0.2 kΩ] — [ideal D2, 0.6 V, 0.2 kΩ] — [ideal D3, 0.6 V, 0.2 kΩ] — **B** (V_o) — 10 kΩ — (+) 3 V (−) — ground, with 5 kΩ from A to ground.



Q3(a) model answer: each diode drawn as an ideal diode, a 0.6 V source (+ towards A) and 0.2 kΩ.

(b) Verify conduction [4]. Assume all three conduct, with current I flowing A → B (work in mA, kΩ, V).

KCL at A:

$$\frac{10 - V_A}{5} = \frac{V_A}{5} + I \;\Rightarrow\; V_A = 5 - 2.5I$$

At B: $V_B = 3 + 10I$.

Diode chain: $V_A - V_B = 3(0.6) + 3(0.2)I = 1.8 + 0.6I$.

Substitute:

$$(5 - 2.5I) - (3 + 10I) = 1.8 + 0.6I \;\Rightarrow\; 0.2 = 13.1I \;\Rightarrow\; I = 0.01527\ \text{mA} = 15.3\ \mu\text{A}$$

$I > 0$ and the same current flows through all three ideal diodes, so **all three are conducting**.

Quick alternative (Thevenin at A: 5 V, 2.5 kΩ): with the diodes removed, the voltage across the string is $5 - 3 = 2$ V. That is more than the 1.8 V needed, so the diodes turn on. Then $I = (5 - 1.8 - 3)/(2.5 + 0.6 + 10) = 15.3$ µA, the same answer.

Check: $V_A = 4.962$ V, $V_B = 3.153$ V, so the chain drops 1.809 V $= 1.8 + 0.6(0.0153)$.

(c) Output [2].

$$V_o = V_B = 3 + 10\ \text{k}\Omega \times 15.3\ \mu\text{A} = \mathbf{3.15\ V}$$

Python: I = 15.267 µA, V_o = 3.1527 V. This agrees with the Test 1 Reference.

Mark breakdown (8):

- a. 1: each diode drawn as ideal diode + 0.6 V + 0.2 kΩ, with correct polarity. 1: the rest of the circuit is correct, including the 3 V source polarity (+ towards the 10 kΩ).
- b. 1: a correct KCL/KVL equation at A, or a correct Thevenin at A. 1: the diode-chain equation including $3r_D I$. 1: solving for I = 15.3 µA. 1: the stated conclusion that I > 0, so all three conduct.
- c. 1: $V_o = 3 + 10I$. 1: 3.15 V.

Most likely ways to lose marks:

- Forgetting the 5 kΩ shunt from A to ground and using 10 V straight into the string. This gives I = (10 − 1.8 − 3)/(15.6) = 0.333 mA, which is wrong.
- Using 0.6 V per diode but leaving out the r_D terms, or writing the 3 V source with the wrong sign. With the wrong sign you get I = 0.473 mA and $V_o = 1.73$ V. Check the sign: a 30-fold jump in current from a sign slip is the clue.

Question 4: Zener regulator design [5 marks]

Paper wording. Design the network below to maintain V_L at 12 V for a load variation (I_L) from 0 mA to 200 mA. That is, determine R_S and V_Z. (Word for word, this is WS3 Q13.)

Q4 figure (from the practice-test paper)

V_Z: the Zener sets the load voltage, so $\mathbf{V_Z = 12\ V}$.

R_S: design at the worst case, full load $I_L = 200$ mA, where the Zener is just in breakdown ($I_Z \to 0$):

$$R_S = \frac{V_i - V_Z}{I_{L,\max} + I_{Z,\min}} \approx \frac{16 - 12}{200\ \text{mA}} = \mathbf{20\ \Omega}$$

Workshop method (WS3 solution, Q13): at full load $R_L = 12/0.2 = 60\ \Omega$. With the Zener drawing nothing, it is a divider:

$$\frac{60 \times 16}{60 + R_S} = 12 \ \Rightarrow \ R_S = 20\ \Omega$$

For the last mark: at $I_L = 0$ the Zener carries all of $I_S = 4/20 = 200$ mA, so it needs $P_{Z,\max} \geq 12 \times 0.2 = 2.4$ W. (R_S dissipates $4 \times 0.2 = 0.8$ W.)

This agrees with the WS3 Q13 solution and the Test 1 Reference.

Mark breakdown (5):

- 1: $V_Z = 12$ V, with the reason given.
- 1: identifying full load (200 mA) as the case that sets R_S, with $I_{Z,\min} \approx 0$.
- 2: the equation (divider or $(V_i - V_Z)/I$) and $R_S = 20\ \Omega$.
- 1: checking the other extreme, $I_L = 0$: $I_Z = 200$ mA and $P_Z = 2.4$ W.

Most likely ways to lose marks:

- Sizing R_S at $I_L = 0$. You cannot get an R_S from that case, and some students divide by a made-up I_Z instead.
- Leaving out the no-load check. The workshop solution stops at 20 Ω, but “design” questions usually reward the Zener current and power rating.

Question 5: series clipper [5 marks]

Paper wording. Determine v_o of the network below for the input shown. (Word for word, this is WS3 Q12.)

Q5 figure (from the practice-test paper)

READING THE FIGURE: v_i is a square wave between +20 V and −5 V. The circuit is v_i(+), then a 4 V battery (− on the input side, + on the diode side), then an ideal diode (cathode

towards the battery, anode towards the output), then a node v_o with 1 kΩ to ground.

Orientation. The only direction current can flow through the diode is from the output node towards the battery. That means current flows *up* through the 1 kΩ, so v_o can only be zero or negative.

Transition voltage (ideal diode, so $v_d = 0$ and $i_d = 0$ at the boundary): $v_i + 4 = 0 \Rightarrow v_{iT} = -4$ V. The diode conducts only for $v_i < -4$ V.

$v_i = +20$ V: the diode is reverse biased (it sees 20 + 4 = 24 V reverse). It is an open circuit, so no current flows and

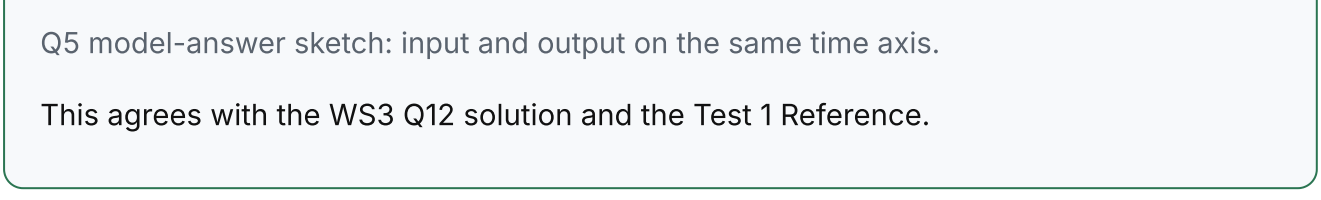
$$v_o = \mathbf{0\ V}$$

$v_i = -5$ V: the input overpowers the 4 V battery, so the diode is on (short). KVL round the loop:

$$+5 - 4 + v_o = 0 \ \Rightarrow \ v_o = \mathbf{-1\ V}$$

Equivalently, $v_o = v_i + 4 = -1$ V, and the current is 1 mA up through the 1 kΩ.

Sketch: a square wave in step with v_i: 0 V while $v_i = 20$ V, and −1 V while $v_i = -5$ V.



Q5 model-answer sketch: input and output on the same time axis.

This agrees with the WS3 Q12 solution and the Test 1 Reference.

Mark breakdown (5):

- 1: transition voltage $v_i = -4$ V, or a correct statement of when the diode conducts.
- 1: $v_i = +20$ V, diode off, $v_o = 0$ V.
- 2: $v_i = -5$ V, diode on, KVL giving $v_o = -1$ V.
- 1: a sketch of v_o against time, aligned with v_i and with levels labelled.

Most likely ways to lose marks:

- Misreading the diode direction and getting $v_o = 20 - 4 = 16$ V for the positive half. Trace the arrow before writing anything.
- Getting the battery sign wrong in the KVL and writing $v_o = -9$ V ($-5 - 4$). The + terminal faces the diode, so it *adds* 4 V to v_i.

Question 6: load line and Q-point [6 marks]

Paper wording. Draw the load line for the network below on the characteristics.
a. Find the Q-point and resulting values I_{C_Q} and V_{CE_Q}.
b. Find the value of β at the Q-point.

Q6 circuit (from the practice-test paper)

Q6 output characteristics (from the practice-test paper)

READING THE FIGURE: emitter-bias circuit with $V_{CC} = 20$ V, $R_B = 270$ kΩ, $R_C = 470\ \Omega$, $R_E = 2.2$ kΩ and $\beta = 125$ printed on the transistor.

Step 1: load line. $V_{CE} = V_{CC} - I_C(R_C + R_E)$ with $I_E \approx I_C$, and $R_C + R_E = 2.67$ kΩ.

- $I_C = 0 \Rightarrow V_{CE} = 20$ V.
- $V_{CE} = 0 \Rightarrow I_C = 20/2.67\text{k} = 7.49$ mA.

Join (20 V, 0) to (0, 7.49 mA) on the graph with a ruler.

Step 2: base current (base loop, which picks the curve):

$$I_{BQ} = \frac{V_{CC} - V_{BE}}{R_B + (\beta + 1)R_E} = \frac{20 - 0.7}{270\text{k} + 126(2.2\text{k})} = \frac{19.3}{547.2\text{k}} = 35.3\ \mu\text{A}$$

Step 3 (a): Q-point. Read where the load line crosses an imagined $I_B \approx 35\ \mu$A curve, halfway between the 30 µA and 40 µA curves. On the printed graph, in the range $V_{CE} \approx 7$–8 V, the 30 µA curve sits at about 4.0 mA and the 40 µA curve at about 5.2 mA. Interpolating to 35.3 µA:

$$I_{CQ} \approx 4.6\ \text{mA (graph)}, \qquad V_{CEQ} \approx 20 - 4.66(2.67) \approx 7.6\ \text{V (graph)}$$

Analytical cross-check using the stated $\beta = 125$:

$$I_{CQ} = 125 \times 35.3\ \mu\text{A} = 4.41\ \text{mA}, \quad I_E = 4.44\ \text{mA}, \quad V_{CEQ} = 20 - 4.41\text{m}(470) - 4.44\text{m}(2200) = 8.15\ \text{V}$$

Either answer is accepted if you show the method: graph values of about 4.6 mA and 7.6 V, or analytical values of 4.41 mA and 8.15 V. The question says “on the characteristics”, so the graph reading is the intended answer. Quote the analytical one as a check. (The E4 pack reads the same graph as about 4.5 mA, 8.0 V, β ≈ 127. Treat **4.5–4.6 mA, 7.6–8.0 V, β ≈ 127–130** as the graph answer.)

Step 3 (b): β at the Q-point.

$$\beta_{dc} = \frac{I_{CQ}}{I_{BQ}} \approx \frac{4.6\ \text{mA}}{35.3\ \mu\text{A}} \approx \mathbf{130}$$

A reading anywhere in 4.5–4.7 mA gives about 128–133.

Where this differs from the Test 1 Reference. The Reference (p. 19) reads $I_{CQ} \approx 4.2$–4.4 mA off the graph and gets $\beta \approx 120$. That graph reading is too low. On the printed characteristics the 30 µA curve is already at about 4.0 mA (it sits on the “4” gridline). The 40 µA curve is at about 5.2 mA. So a 35 µA operating point cannot be at 4.2 mA. It is about 4.6–4.7 mA, which gives $\beta \approx 130$ and $V_{CEQ} \approx 7.6$ V. I checked this by measuring the curve positions on the rasterised page: 30 µA gives 4.03 mA and 40 µA gives 5.22 mA at $V_{CE} = 7.5$ V. The Reference’s analytical values (4.41 mA, 8.15 V) are correct for $\beta = 125$. The difference simply shows that the printed device has $\beta \approx 130$, a little above the stated 125.

Mark breakdown (6):

- 1: load-line equation and both intercepts (20 V and 7.49 mA).
- 1: load line drawn correctly on the characteristics.
- 1: $I_{BQ} = 35.3\ \mu$A, with the $(\beta + 1)R_E$ term.
- 2: the Q-point marked, and I_{CQ} and V_{CEQ} read off (about 4.6 mA and 7.6 V; analytical values of 4.41 mA and 8.15 V with working are also accepted).
- 1: $\beta = I_{CQ}/I_{BQ}$, using your *graph* values (about 130).

Most likely ways to lose marks:

- Writing $I_B = 19.3/270\text{k} = 71.5\ \mu$A, which forgets that R_E is reflected into the base loop as $(\beta + 1)R_E$.
- Using only R_C for the load line ($I_{C,sat} = 42.6$ mA) and forgetting R_E. Or, in (b), quoting $\beta = 125$ back instead of dividing your graph readings.

Question 7: Thevenin at the base [10 marks]

Paper wording. For the circuit shown on the right, $\beta = 50$, $V_{BE,on} = 0.7$ V, and $V_{CE,sat} = 0.2$ V. Answer the following questions assuming that the BJT operates in the **active state**.
a) Find the Thevenin equivalent circuit (i.e., V_{Th} and R_{Th}) seen from the base of the transistor. [4 marks]
b) Redraw the circuit by replacing the transistor with its **DC equivalent circuit** model and Thevenin equivalent model in the base. [4 marks]
c) Determine collector current (I_C) and voltage (V_C). [2 marks]

Q7 figure (from the practice-test paper)

(a) Thevenin [4].

Resistance: zero both sources (short them to their rails):

$$R_{Th} = R_1 \parallel R_2 = \frac{15 \times 30}{45} = \mathbf{10\ k\Omega}$$

Voltage: open circuit at the base, by superposition:

$$V_{Th} = 9 \cdot \frac{30}{45} + (-12) \cdot \frac{15}{45} = 6 - 4 = \mathbf{2\ V}$$

Check: the node sits 30/45 of the way from −12 V to +9 V, which is $-12 + 21(2/3) = 2$ V.

(b) Redraw [4]. The base side is $V_{Th} = 2$ V in series with $R_{Th} = 10$ kΩ, feeding the base. Between B and E (ground) there is a 0.7 V source (+ at B). Between C and E there is a dependent current source $\beta I_B = 50 I_B$, pointing C → E. Above it, $R_C = 1.5$ kΩ goes to $V_{CC} = 12$ V. Label I_B, I_C and V_C.

Q7(b) model answer: Thévenin base source with the active-mode DC model.

(c) I_C and V_C [2]. Base loop (emitter grounded, so there is no R_E term):

$$I_B = \frac{V_{Th} - V_{BE}}{R_{Th}} = \frac{2 - 0.7}{10\text{k}} = 130\ \mu\text{A}$$

$$I_C = \beta I_B = 50 \times 130\ \mu\text{A} = \mathbf{6.5\ mA}$$

$$V_C = V_{CC} - I_C R_C = 12 - 6.5\text{m} \times 1.5\text{k} = \mathbf{2.25\ V}$$

Check: $V_{CE} = V_C = 2.25\ \text{V} > 0.2$ V and $I_B > 0$, so the transistor is active as assumed.

This agrees with the Test 1 Reference.

Mark breakdown (10):

- a. 2: R_{Th} (method of zeroing the sources, then 10 kΩ). 2: V_{Th} (superposition or divider between the two rails, then 2 V).
- b. 1: base Thevenin source and resistor. 1: $V_{BE} = 0.7$ V source with correct polarity. 1: βI_B current source C → E, with correct direction. 1: R_C and V_{CC}, emitter to ground, and labels.
- c. 1: I_B and $I_C = 6.5$ mA. 1: $V_C = 2.25$ V, with the active-state check.

Most likely ways to lose marks:

- Treating −12 V as ground and getting $V_{Th} = 9 \times 30/45 = 6$ V. This also gives $I_C = 26.5$ mA and an impossible V_C. Always include both rails.

- In (b), drawing the current source in the wrong direction or as an independent source, or leaving out the 0.7 V source. These are easy marks that are often lost.

Question 8: two transistors [6 marks]

Paper wording. Consider the circuit below where Q_1 and Q_2 have different current gain, $\beta_1 = 100$ and $\beta_2 = 50$. You can assume $V_{BE,on}$ = 0.7 V and $V_{CE,sat} = 0.2$ V for both BJTs and both BJTs are in **active state**. Find the output voltage, V_{C2} by showing your work.

Q8 figure (from the practice-test paper)

READING THE FIGURE: Q_1 (NPN) has its base and collector tied together at 2.5 V. Its emitter drives the emitter of Q_2 (PNP). Q_2's base goes through 10 kΩ to a 1 V terminal. Its collector is V_{C2}, with 1 kΩ to ground.

Step 1: Q_1 sets a level. Q_1 is diode-connected (B tied to C), so it simply drops one V_{BE}:

$$V_{E1} = 2.5 - 0.7 = 1.8 \text{ V} = V_{E2}$$

Step 2: Q_2 base. Q_2 is a PNP in active mode, so its emitter is 0.7 V above its base:

$$V_{B2} = 1.8 - 0.7 = 1.1 \text{ V}$$

Step 3: base current. Current flows out of the base, through 10 kΩ, to the 1 V terminal. That is the correct direction for a PNP.

$$I_{B2} = \frac{1.1 - 1}{10 \text{ k}\Omega} = 10 \ \mu\text{A}$$

Step 4: collector.

$$I_{C2} = \beta_2 I_{B2} = 50 \times 10 \ \mu\text{A} = 0.5 \text{ mA} \ (\text{out of the collector, down through } 1 \text{ k}\Omega)$$

$$V_{C2} = 0.5 \text{ mA} \times 1 \text{ k}\Omega = \mathbf{0.5 \text{ V}}$$

> **CHECK:** $V_{EC2} = 1.8 - 0.5 = 1.3 \text{ V} > 0.2$ V and $I_{B2} > 0$, so Q_2 is active. $I_{E2} = 51 \times 10 \ \mu\text{A} = 0.51$ mA, which Q_1 supplies. β_1 does not affect V_{C2}, because Q_1 only fixes the 1.8 V level. Write this down, because it shows you understood the circuit.

This agrees with the Test 1 Reference.

Mark breakdown (6):

- 1: recognising Q_1 as diode-connected, so $V_{E1} = V_{E2} = 1.8$ V.
- 1: $V_{B2} = 1.1$ V, using the PNP polarity.
- 1: $I_{B2} = 10$ μA, with the correct direction.
- 1: $I_{C2} = 0.5$ mA.
- 1: $V_{C2} = 0.5$ V.
- 1: active-state check ($V_{EC2} = 1.3$ V > 0.2 V), or the comment that β_1 is irrelevant.

Most likely ways to lose marks:

- Using NPN polarity on Q_2 ($V_{B2} = 1.8 + 0.7 = 2.5$ V). This gives $I_B = 150$ μA into the base, and the whole answer goes wrong.
- Trying to push β_1 into the calculation, for example by computing I_{B1} and calling it I_{B2}. If Q_1's numbers appear anywhere in your V_{C2} line, check again.

After marking

Q	Marks	Your score	Topic	Revise in
1 (I)	1		pn junction, depletion region	E1
1 (II)	1		dynamic resistance r_d	E2
1 (III)	1		full-wave vs half-wave rectification	E3
1 (IV)	1		BJT physics (thin base, α, β)	E4
1 (V)	1		DC vs AC (small-signal) analysis	E4 / E6
2	5		exponential (Shockley) model, $\Delta V = nV_T \ln$ ratio	E2
3	8		third piecewise-linear diode model, verify conduction	E2 (Thevenin step: E1)
4	5		Zener regulator design	E3
5	5		series clipper with a battery	E3
6	6		emitter bias, load line, Q-point from characteristics	E4 / E5

Q	Marks	Your score	Topic	Revise in
7	10		Thevenin at the base, active-mode DC model	E5 (Thevenin: E1)
8	6		multi-transistor DC, PNP polarity	E4 / E5

How to read your score. If you scored 40 or more, you are on track. Redo only the questions where you lost marks, a day before Test 1. If you scored 30–39, do the "Revise in" pack for every question where you lost 2 or more marks. If you scored under 30, prioritise Q3, Q6 and Q7 (24 marks between them, all method marks) and their packs, E2 and E5.

Quick recall check

1. Two decades of diode current at $n = 1.5$, 27 °C: how much does V_D change?
2. In a Zener regulator, at which load current do you size R_S, and which case do you use to check the Zener power rating?
3. For an emitter-bias circuit, what resistance does R_E look like from the base loop?
4. In a series clipper with an ideal diode and a battery, how do you find the transition voltage?
5. A PNP in active mode has its emitter at 1.8 V. What is its base voltage, and which way does the base current flow?

1. $\Delta V = nV_T \ln 100 = 1.5 \times 25.9\ \text{mV} \times 4.605 \approx 179$ mV, which is about 89 mV per decade.
2. Size R_S at maximum load current ($I_Z \approx 0$): $R_S = (V_i - V_Z)/I_{L,\max}$. Check the power at no load, where the Zener takes all of I_S: $P_Z = V_Z I_S$.
3. $(\beta + 1)R_E$, so $I_B = (V_{CC} - V_{BE})/(R_B + (\beta + 1)R_E)$.
4. Set the diode at the boundary ($v_d = 0$, $i_d = 0$) and write KVL round the input loop. Here $v_i + 4 = 0$ gives $v_{iT} = -4$ V.
5. $V_B = 1.8 - 0.7 = 1.1$ V. Base current flows *out* of the base, and I_C flows *out* of the collector.